Item 1

a)- The walls were covered with soft, thick woollen materials to reduce the effect of reverberation / reflection (echoes) because they able absorb the sound waves. ~~02~~ 03

- If it were left bare and hard, the incident sound waves would mix with the reflected wave, producing a prolonged and confused sound.

b)- The appearance of an object depends on the light falling on it and that reflected or absorbed by the object.

- When green light falls on the std's uniform, the blue skirt appears black, the white skirt appears green and the yellow collar appears green. 04

- When the red light falls on the std's uniform, the blue skirt appears black, the white shirt appears red and the yellow collar appears red.

c) Number of images $= \frac{360}{\theta} - 1$

$= \frac{360}{60} - 1$ 03

$= 5$ images

A friend saw 5 images while in the studio.

d) $V = \lambda f$

$3 \times 10^8 = 3.3 \times f$ $1MHz = 10^6 Hz$

$f = 90{,}909{,}090.9$ Hz 05

$= \left(\frac{90{,}909{,}090.9}{10^6}\right)$ MHz

$f = 90.9$ MHz

They went to KT FM since it was broadcasting at 90.9MHz.

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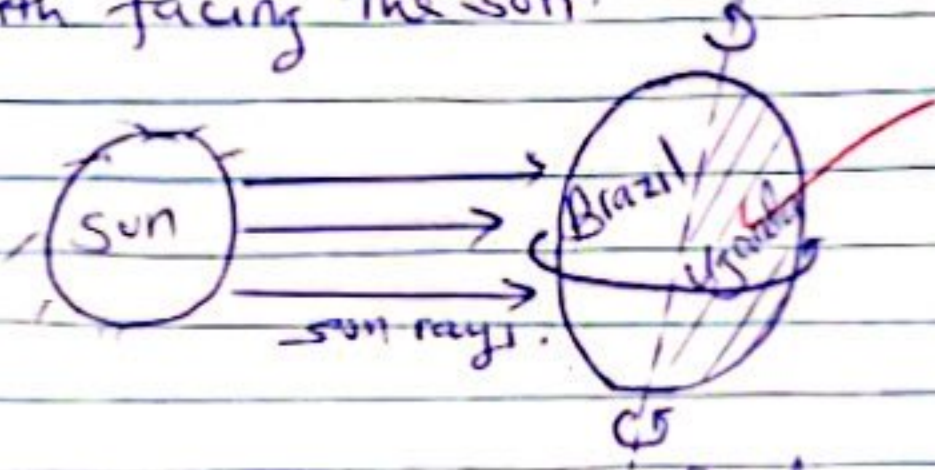
8/15

Item 2.

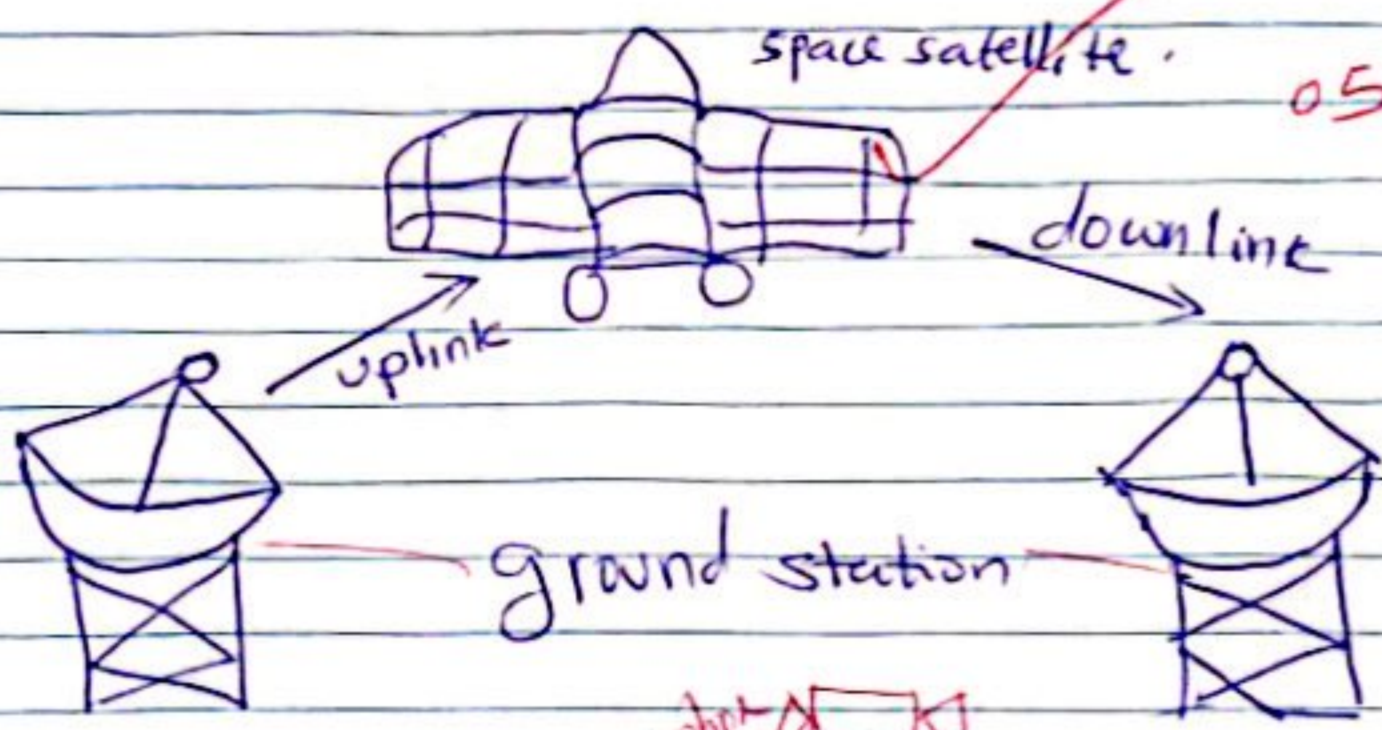
a) Day and night occur due to the rotation of the earth on its fixed vertical axis
One complete rotation takes 24 hours. 1 spherical shape
As the earth rotates, the hemisphere/part facing the sun receives light and heat hence experiences day time. while the hemisphere/part of the earth facing the opposite side of the sun receives no light hence experiences night time.
Therefore Uganda was in the part of the earth facing away from the sun while Brazil was in the part of the earth facing the sun. 05

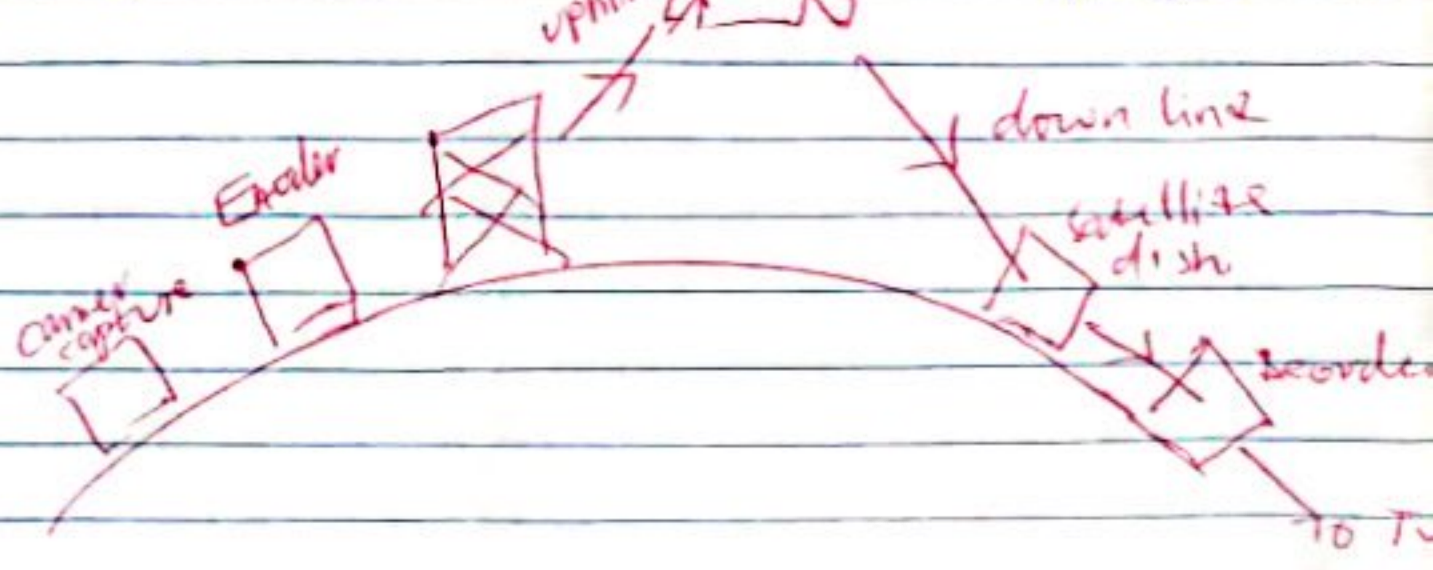


b)
Seasons occur due to the revolution of the earth around the sun/its orbit and tilting of the earth.
One complete revolution of the earth is equivalent to 365 days
The earth is tilted at on its axis at an angle of 23.5°
The earth is elliptical in shape implying that different parts of the earth receive different amount of sunlight throughout the year. 05
The hemisphere that is tilted maximally towards the sun receives alot of heat/sunlight experiencing high temperatures hence summer season.
The hemisphere that is tilted maximally away from the sun receives less/no heat experiencing low temperatures hence winter season hence Brazil was ~~facing~~ experiencing winter season (relevant figure).

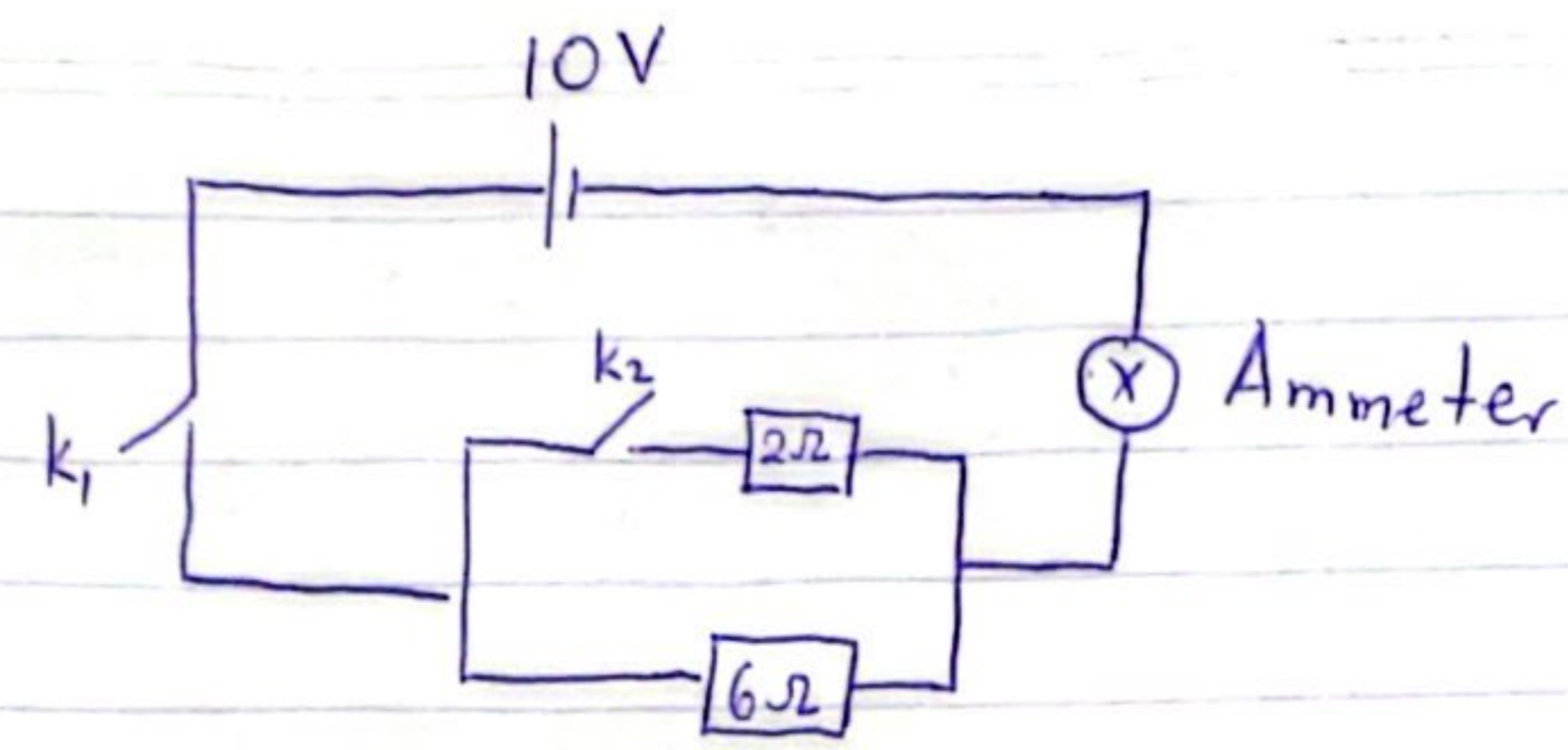
c) John was able to watch matches in Brazil due to the use of Satellite communication.
DsTv providers use cameras to capture pictures turning them into electronic signals.
DsTv providers send these signals from ground stations inform of radio waves through the uplink channel to the geostationary satellite (space satellite).
These signals are weak due to long distance travelled.
Space satellite amplifies / boosts their strength and then re-transmit them to the satellite dish installed in the subscriber's home.
These signals are then sent to the decoder which converts them into audio, video, images, and voices

05





5 (a)



When k_1 is closed and k_2 open

$V = IR$

$10 = I \times 6$

$I = 1.667A$

When k_1 is open and k_2 closed.

$I = 0A$ No current flows in the circuit

When k_1 is closed and k_2 closed

$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

$\frac{1}{R} = \frac{1}{2} + \frac{1}{6}$

$R = 1.5\Omega$

$V = IR$

$10 = I \times 1.5$

$I = 6.667A$

08

Conclusions

For a large current through the ammeter at X by closing one switch, then switch k_1 should be closed.

For maximum current by opening and closing any switch, then both switches should be closed.

02

(b) i) Increase in temperature causes an increase in resistance hence the current through the ammeter reduces

This is because higher temperatures cause atoms in the wire to vibrate more vigorously reducing the drift velocity of the electrons

03

~~iii)~~ ~~Temp↑ → resist~~ $T\uparrow \rightarrow R\uparrow \rightarrow I\downarrow$

ii) Increasing the crosssectional area reduces the resistance of the wires and this increases the current through the ~~ma~~ ammeter

This is because more space for electrons to flow is provided.

$A\uparrow \rightarrow R\downarrow \rightarrow I\uparrow$

03

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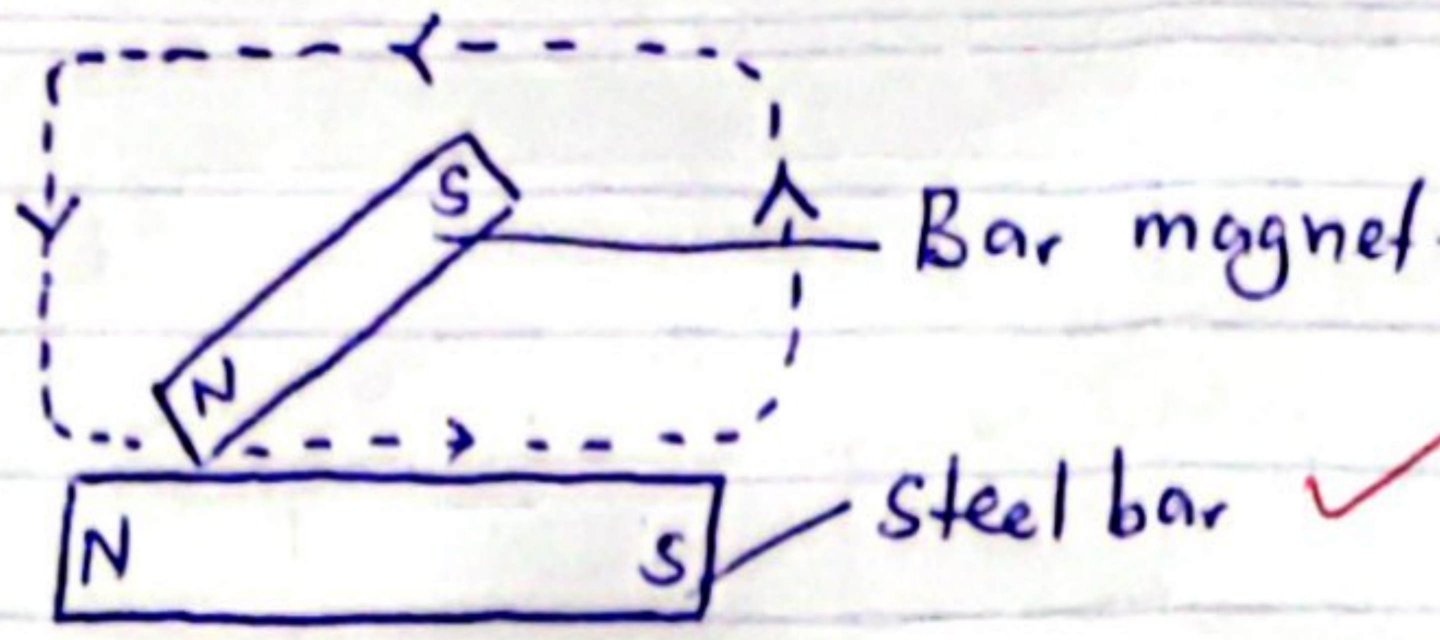
(c)



- A steel bar to be magnetised is placed on a table.
- The steel bar is stroked from end to end repeatedily with the same pole of the bar magnet in the same direction.
- The bar magnet is raised each time it reaches the end of the steel bar.
- The pole at the end of the steel bar where the stroking ends is opposite to that of the stroking pole. 04

/20

Item three

To refuel every after 210km

(a) - The jerking was caused by inertia when the vehicle accelerated or decelerated.

- Inertia is the natural tendency of objects in motion to stay in motion and objects at rest to stay at rest, unless a force causes its velocity to change.

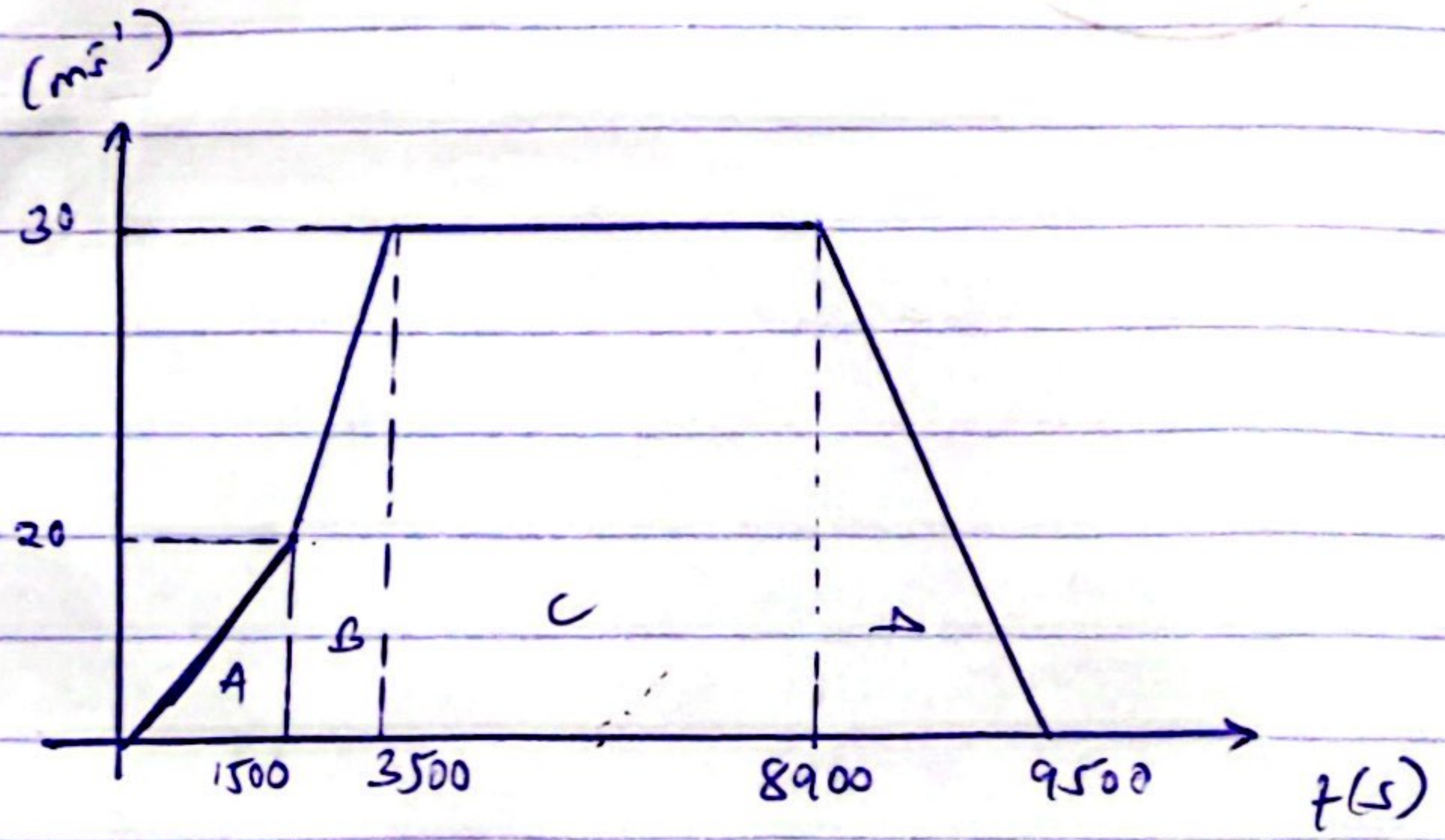
02

(b) The car remained cold because of the air conditioning system

- A refrigerant liquid inside the AC evaporates
- During evaporation, it absorbs heat from the air inside the car.
- The air blown back becomes cooler making the interior of the car to stay cool.

03

(c)



Total distance = Total area under the graph

Area A

Area = $\frac{1}{2} \times b \times h$

$= \frac{1}{2} \times 1500 \times 20$

$= 15,000m$

Area B

Area $= \frac{1}{2} h (a+b)$

$= \frac{1}{2} \times 2000 (20+30)$

$= 50,000 m$

Area C

Area $= L \times W$

$= 5400 \times 30$

$= 162,000 m$

Area D

Area $= \frac{1}{2} bh$

$= \frac{1}{2} \times 600 \times 30$

$= 9000 m$

Total distance $= 15000 + 50,000 + 162,000 + 9000$

$= 236,000 m$

from KHDMDCM

1m $= \frac{1}{1000}$ km

$236,000m = (\frac{1}{1000} \times 236000)$ km

$= 236 km$

Since 236 km > 210 km, the vehicle must have been fueled.

Advice → The vehicle was refueled, because the distance travelled exceeded 210 km.

11

(d)

concept	
Inertia	Seat belts in cars prevent passengers from moving forward during sudden stops
Velocity time graphs	Used in vehicle tracking systems and transport monitoring.
Acceleration and deceleration	Used in vehicles in designing safe braking systems.

04

any two

20/20

Provided - Two pulleys

θ_2 = 59°C

g = 10 ms^{-2}

HC = 102 JK^{-1}

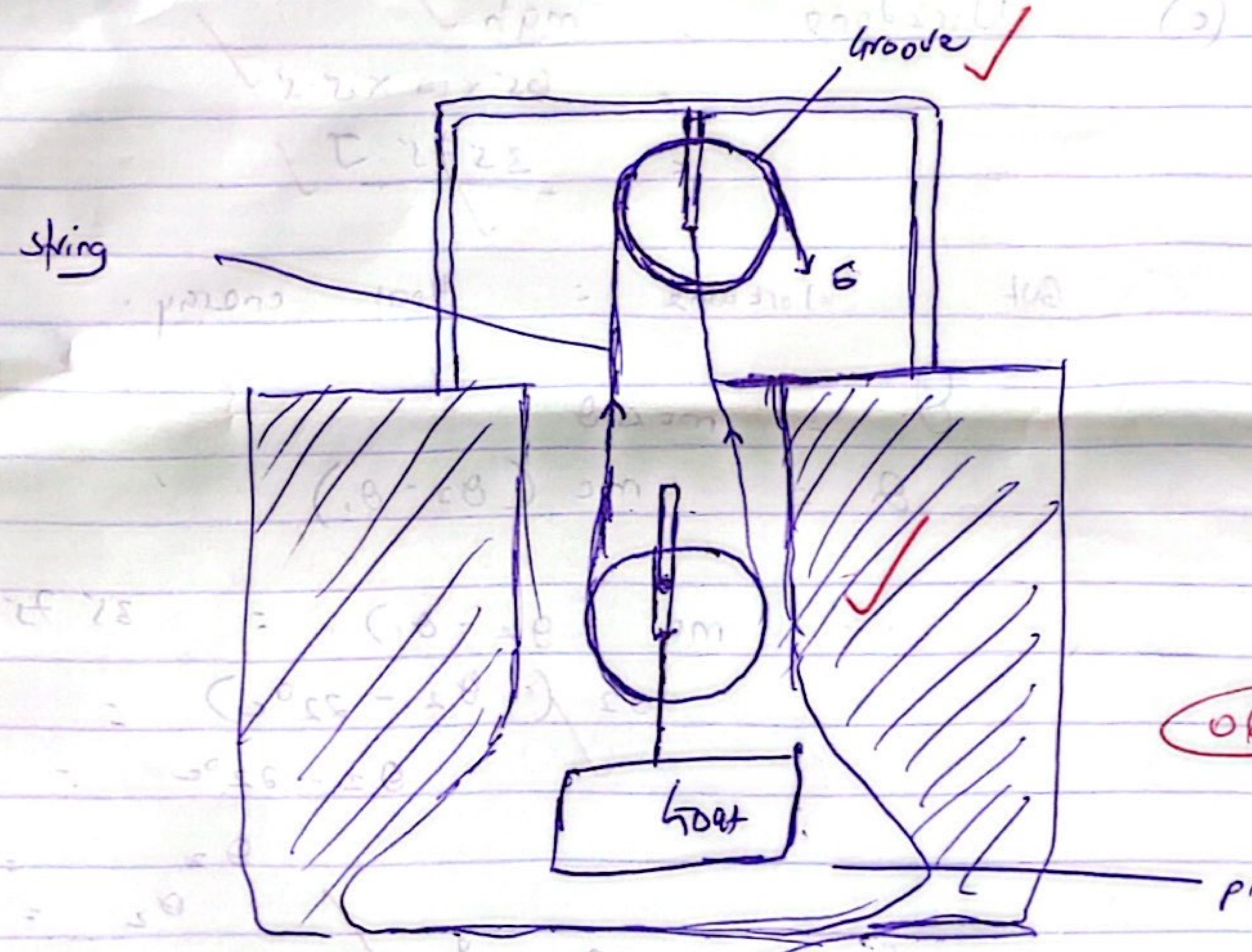
h = 5.5 m

m = 65 kg

θ_1 = 22°C

(a) Procedure

- Fix one pulley at the top of the pit (fixed pulley)
- Attach one pulley to the goat (movable pulley)
- Pass the rope through both pulleys.



(06)

- This gives m.a of two and it implies that the effort is reduced by a half and therefore the worker lifts with less force.

(b) - Some heat will be generated due to the friction between the rope and the groove.

- The heat can be reduced through oiling the groove.
- It can also reduced by using a light smooth rope and smooth groove

(03)

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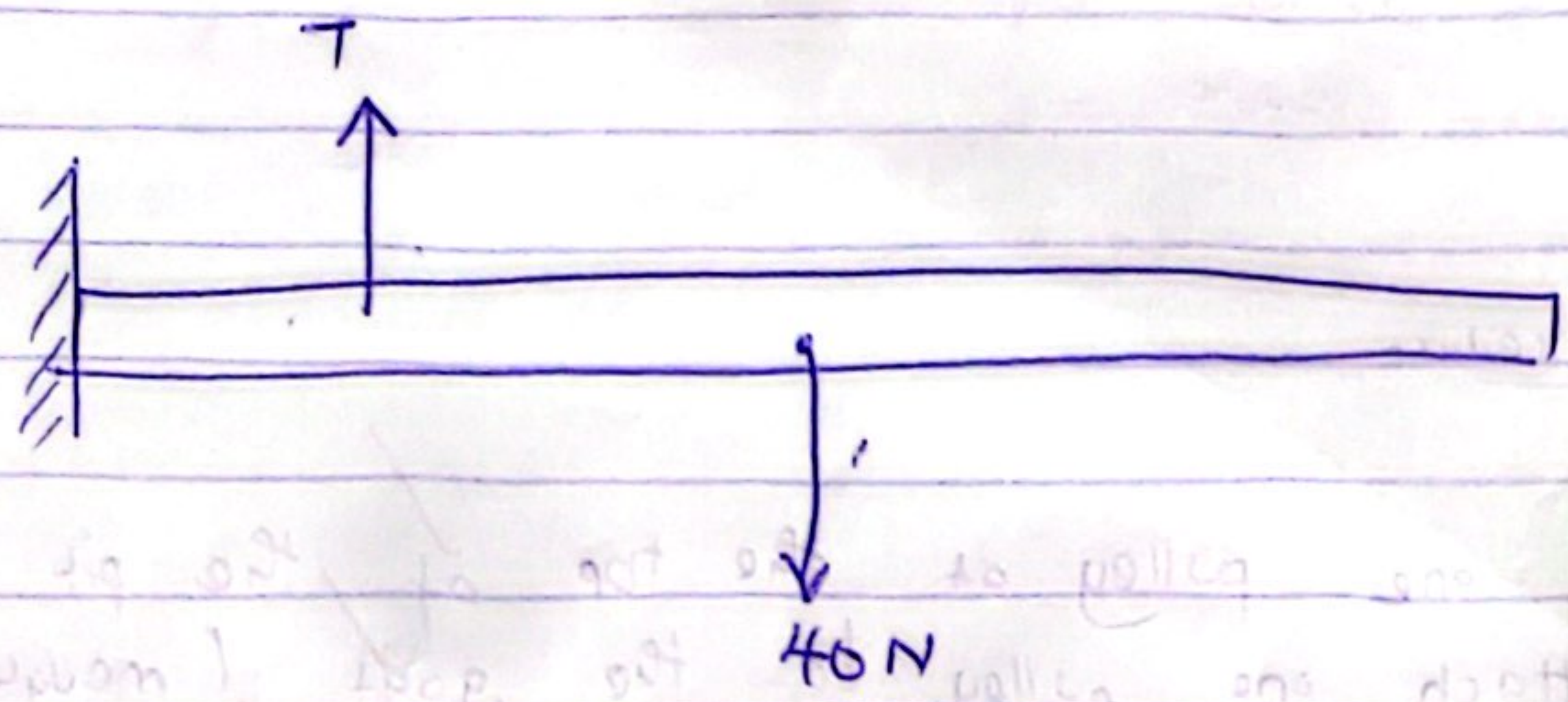

$C = \frac{Q}{\Delta\theta}$

$c = \frac{Q}{m\Delta\theta}$

$c = \frac{C}{m}$

$C = mc$

(c) Workdone $= mgh$

$= 65 \times 10 \times 5.5$

$= 3575\ J$

But Workdone = Heat energy.

11

$Q = mc\Delta\theta$

$Q = mc(\theta_2 - \theta_1)$

$\therefore mc(\theta_2 - \theta_1) = 3575$

$102(\theta_2 - 22^\circ c) = 3575$

$\theta_2 - 22^\circ c = 35.05$

$\theta_2 = 22 + 35.05$

$\theta_2 = 57.05^\circ c$

Since $57.05^\circ C < 59^\circ c$

The string is safe and suitable for use

20/20